

Q1. Load the given retail sales dataset into a Pandas DataFrame and display the first 5 rows along with the dataset information to understand its structure

```
import pandas as pd.py 1 X Welcome Extension: C# Dev Kit
C: > Users > Inbox > OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd.py > ...
1 import pandas as pd
2
3 df = pd.read_csv(r"C:\Users\Inbox\Downloads\1232.csv")
4
5 # First 5 rows
6 print(df.head())
7
8 # Dataset information
9 print(df.info())
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

FileNotFoundError: [Errno 2] No such file or directory: 'C:\\Users\\Inbox\\Downloads\\retail_dataset.csv'

PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code\ & C:\Users\Inbox\AppData\Local\Microsoft\WindowsApps\python3.13.exe "c:/Users/Inbox/OneDrive/Desktop/DATA ANALYTICS/Python/import pandas as pd.py"

	Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region
0	T001	C015	Books	7.0	167.0	02-02-2023	North
1	T002	C004	Toys	4.0	374.0	17-02-2023	South
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South
4	T005	C022	Books	7.0	282.0	27-07-2024	West

<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
Column Non-Null Count Dtype
--- ---
0 Transaction ID 100 non-null str
1 Customer ID 100 non-null str
2 Product Category 97 non-null str
3 Quantity 98 non-null float64
4 Price 95 non-null float64
5 Date 100 non-null str
6 Region 98 non-null str
dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code\

Q2. Inspect the dataset and identify missing values in each column. Display the total number of missing values present in the dataset.

```
9 print(df.info())
10
11 print(df.isnull().sum())
12
13 total_missing = df.isnull().sum().sum()
14 print("Total Missing Values:", total_missing)
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South
4	T005	C022	Books	7.0	282.0	27-07-2024	West

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5 Date 100 non-null str
6 Region 98 non-null str
dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
Transaction ID 0
Customer ID 0
Product Category 3
Quantity 2
Price 5
Date 0
Region 2
dtype: int64
Total Missing Values: 12
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code\

Q3. Replace missing values in the Quantity and Price columns with their respective mean values using Pandas.

```
C:\Users\Inbox> OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd.py > ...
13 total_missing = df.isnull().sum().sum()
14 print("Total Missing Values:", total_missing)
15
16 df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
17 df['Price'] = df['Price'].fillna(df['Price'].mean())
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
1 T002 C004 Toys 4.0 374.0 17-02-2023 South
2 T003 C033 Furniture 8.0 1059.0 17-05-2024 South
3 T004 C047 Groceries 5.0 304.0 11-02-2024 South
4 T005 C022 Books 7.0 282.0 27-07-2024 West
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dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
Transaction ID 0
Customer ID 0
Product Category 3
Quantity 2
Price 5
Date 0
Region 2
dtype: int64
Total Missing Values: 12
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>

Q4. Remove all rows where the Product Category or Region column contains missing values.

```
View Go Run Terminal Help < -> Search
import pandas as pd.py 1 x Welcome Extension: C# Dev Kit
C:\Users\Inbox> OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd.py > ...
15
16 df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mean())
17 df['Price'] = df['Price'].fillna(df['Price'].mean())
18
19 df = df.dropna(subset=['Product Category', 'Region'])
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
1 T002 C004 Toys 4.0 374.0 17-02-2023 South
2 T003 C033 Furniture 8.0 1059.0 17-05-2024 South
3 T004 C047 Groceries 5.0 304.0 11-02-2024 South
4 T005 C022 Books 7.0 282.0 27-07-2024 West
```

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dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
Transaction ID 0
Customer ID 0
Product Category 3
Quantity 2
Price 5
Date 0
Region 2
dtype: int64
Total Missing Values: 12
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>

Q5. Create a new column called Revenue using NumPy, where: $\text{Revenue} = \text{Quantity} \times \text{Price}$
Display the updated dataset with the new column.

```
import pandas as pd,py 2 X Welcome Extension: C# Dev Kit
C: > Users > Inbox > OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd,py > ...
21 import numpy as np
22
23 df['Revenue'] = np.multiply(df['Quantity'], df['Price'])
24
25 print(df.head())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	Transaction ID	100 non-null	str
1	Customer ID	100 non-null	str
2	Product Category	97 non-null	str
3	Quantity	98 non-null	float64
4	Price	95 non-null	float64
5	Date	100 non-null	str
6	Region	98 non-null	str

dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
Transaction ID 0
Customer ID 0
Product Category 3
Quantity 2
Price 5
Date 0
Region 2
dtype: int64
Total Missing Values: 12

Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region	Revenue	
0	T001	C015	Books	7.0	167.0	02-02-2023	North	1169.0
1	T002	C004	Toys	4.0	374.0	17-02-2023	South	1496.0
2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South	8472.0
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South	1520.0
4	T005	C022	Books	7.0	282.0	27-07-2024	West	1974.0

PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>

Q6. Using NumPy, calculate the total revenue generated from all transactions in the dataset.

```
import pandas as pd,py 2 X Welcome Extension: C# Dev Kit
C: > Users > Inbox > OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd,py > ...
24 problems in this file
25 print(df.head())
26
27 total_revenue = np.sum(df['Revenue'])
28 print(total_revenue)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Column Non-Null Count Dtype

#	Column	Non-Null Count	Dtype
0	Transaction ID	100 non-null	str
1	Customer ID	100 non-null	str
2	Product Category	97 non-null	str
3	Quantity	98 non-null	float64
4	Price	95 non-null	float64
5	Date	100 non-null	str
6	Region	98 non-null	str

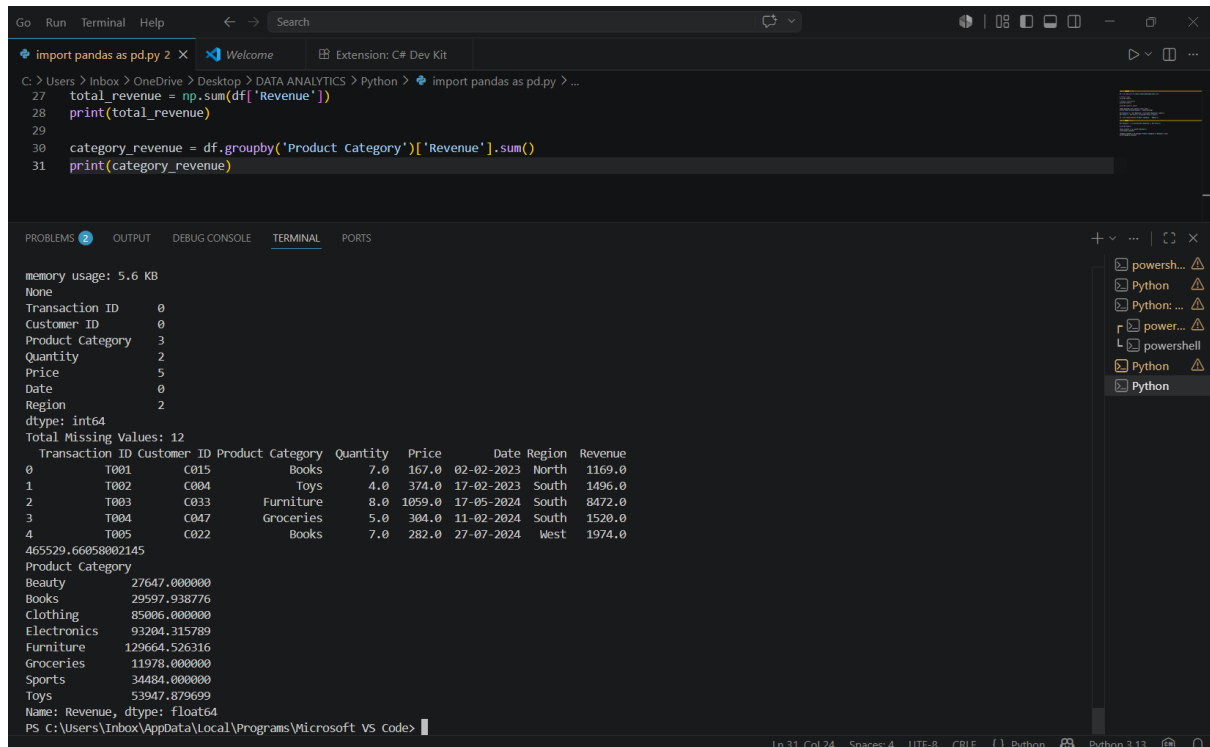
dtypes: float64(2), str(5)
memory usage: 5.6 KB
None
Transaction ID 0
Customer ID 0
Product Category 3
Quantity 2
Price 5
Date 0
Region 2
dtype: int64
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Transaction ID	Customer ID	Product Category	Quantity	Price	Date	Region	Revenue	
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2	T003	C033	Furniture	8.0	1059.0	17-05-2024	South	8472.0
3	T004	C047	Groceries	5.0	304.0	11-02-2024	South	1520.0
4	T005	C022	Books	7.0	282.0	27-07-2024	West	1974.0

465529.66058002145
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>

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Q7. Group the dataset by Product Category and calculate the total revenue generated by each category.



The screenshot shows a VS Code editor with a Python file named 'import pandas as pd.py'. The code in the file is as follows:

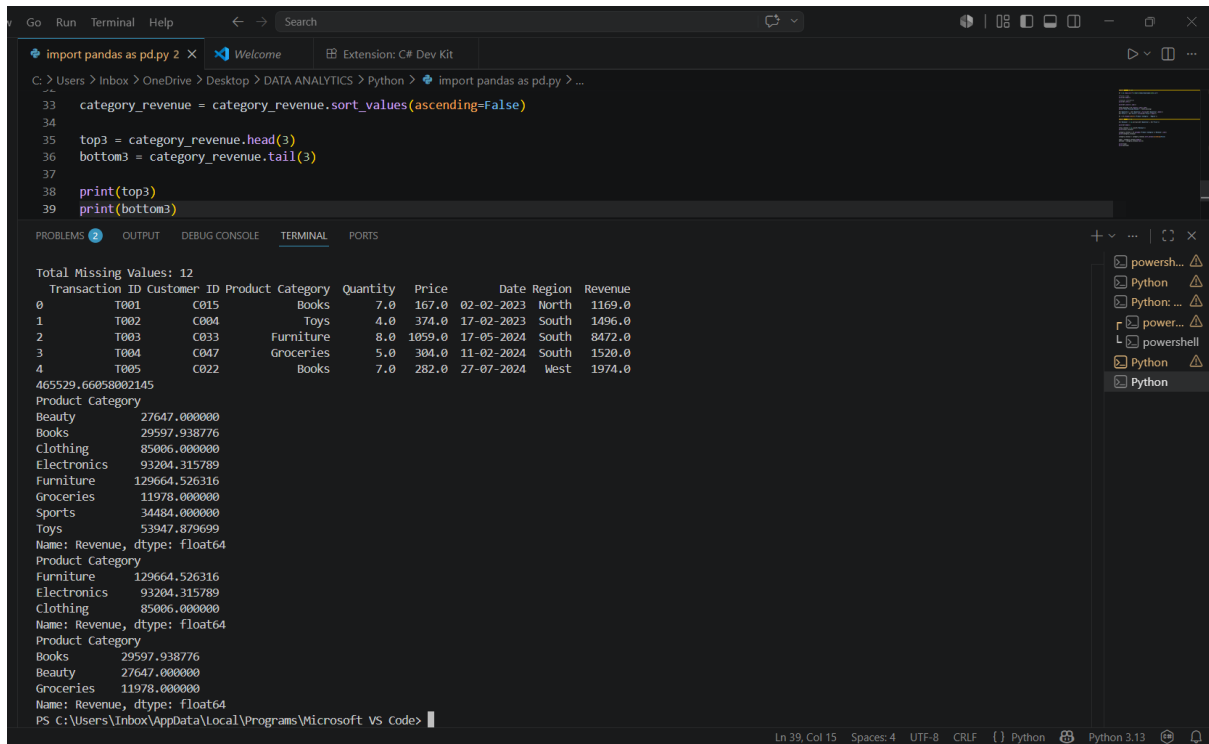
```
27 total_revenue = np.sum(df['Revenue'])
28 print(total_revenue)
29
30 category_revenue = df.groupby('Product Category')['Revenue'].sum()
31 print(category_revenue)
```

The terminal output shows the following information:

```
memory usage: 5.6 KB
None
Transaction ID      0
Customer ID         0
Product Category    3
Quantity            2
Price               5
Date                0
Region              2
dtype: int64
Total Missing Values: 12
Transaction ID Customer ID Product Category Quantity Price Date Region Revenue
0      T001      C015      Books              7.0   167.0  02-02-2023  North   1169.0
1      T002      C004      Toys               4.0   374.0  17-02-2023  South   1496.0
2      T003      C033      Furniture          8.0  1059.0  17-05-2024  South   8472.0
3      T004      C047      Groceries          5.0   304.0  11-02-2024  South   1520.0
4      T005      C022      Books              7.0   282.0  27-07-2024  West    1974.0
465529.66058002145
Product Category
Beauty      27647.000000
Books       29597.938776
Clothing    85006.000000
Electronics 93204.315789
Furniture   129664.526316
Groceries   11978.000000
Sports      34484.000000
Toys        53947.879699
Name: Revenue, dtype: float64
```

The status bar at the bottom indicates the file is 'Python 3.13' and the terminal is running 'Python 3.13'.

Q8. Based on the revenue calculated for each product category, identify the top 3 categories with the highest revenue and the bottom 3 categories with the lowest revenue.

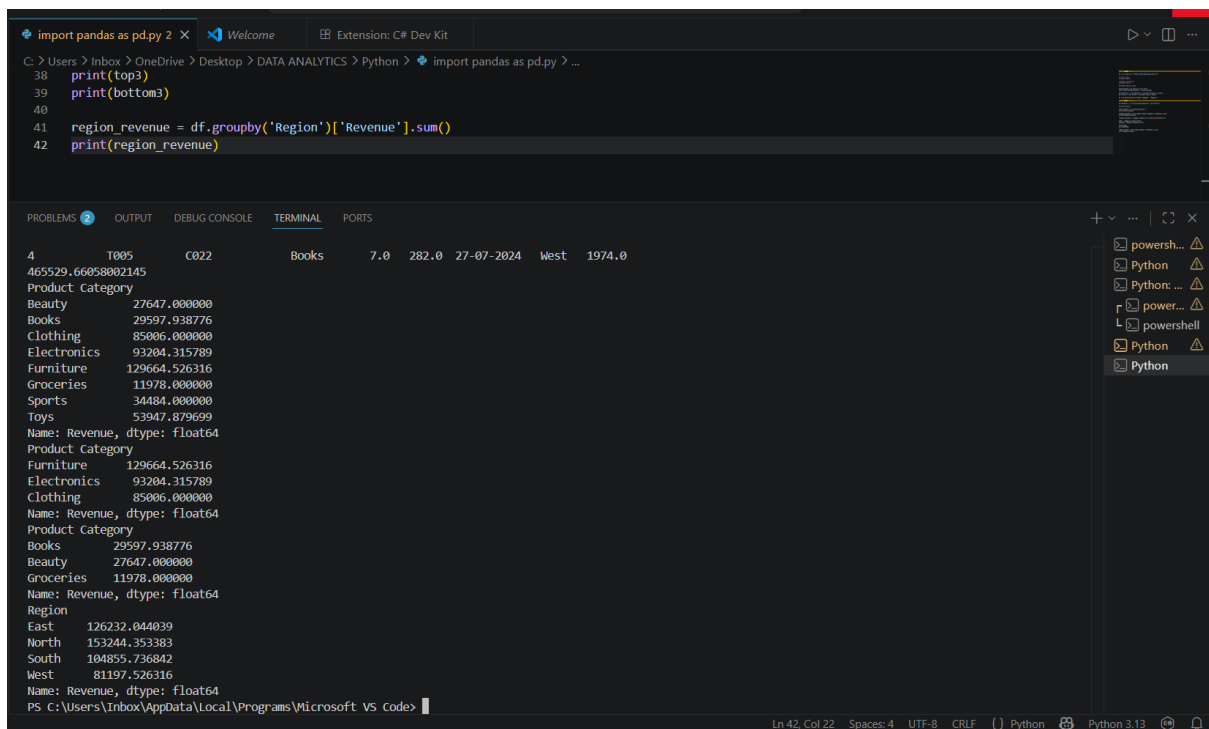


```
import pandas as pd
category_revenue = category_revenue.sort_values(ascending=False)
top3 = category_revenue.head(3)
bottom3 = category_revenue.tail(3)
print(top3)
print(bottom3)
```

Terminal Output:

```
Total Missing Values: 12
Transaction ID Customer ID Product Category Quantity Price Date Region Revenue
0 T001 C015 Books 7.0 167.0 02-02-2023 North 1169.0
1 T002 C004 Toys 4.0 374.0 17-02-2023 South 1496.0
2 T003 C033 Furniture 8.0 1059.0 17-05-2024 South 8472.0
3 T004 C047 Groceries 5.0 304.0 11-02-2024 South 1520.0
4 T005 C022 Books 7.0 282.0 27-07-2024 West 1974.0
465529.66058002145
Product Category
Beauty 27647.000000
Books 29597.938776
Clothing 85006.000000
Electronics 93204.315789
Furniture 129664.526316
Groceries 11978.000000
Sports 34484.000000
Toys 53947.879699
Name: Revenue, dtype: float64
Product Category
Furniture 129664.526316
Electronics 93204.315789
Clothing 85006.000000
Name: Revenue, dtype: float64
Product Category
Books 29597.938776
Beauty 27647.000000
Groceries 11978.000000
Name: Revenue, dtype: float64
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>
```

Q9. Group the dataset by Region and calculate the total revenue generated in each region. Identify the region with the highest revenue and the region with the lowest revenue.



```
print(top3)
print(bottom3)
region_revenue = df.groupby('Region')['Revenue'].sum()
print(region_revenue)
```

Terminal Output:

```
4 T005 C022 Books 7.0 282.0 27-07-2024 West 1974.0
465529.66058002145
Product Category
Beauty 27647.000000
Books 29597.938776
Clothing 85006.000000
Electronics 93204.315789
Furniture 129664.526316
Groceries 11978.000000
Sports 34484.000000
Toys 53947.879699
Name: Revenue, dtype: float64
Product Category
Furniture 129664.526316
Electronics 93204.315789
Clothing 85006.000000
Name: Revenue, dtype: float64
Product Category
Books 29597.938776
Beauty 27647.000000
Groceries 11978.000000
Name: Revenue, dtype: float64
Region
East 126232.044039
North 153244.353383
South 104855.736842
West 81197.526316
Name: Revenue, dtype: float64
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>
```

Q10. Extract the month from the Date column using Pandas and group the dataset by month to calculate the total monthly revenue. Also, calculate the mean, median, and standard deviation of the Revenue column using NumPy

```
import pandas as pd.py 2 X Welcome Extension: C# Dev Kit
C: > Users > Inbox > OneDrive > Desktop > DATA ANALYTICS > Python > import pandas as pd.py > ...
54 std_rev = np.std(df['Revenue'])
55
56 print("Mean:", mean_rev)
57 print("Median:", median_rev)
58 print("Standard Deviation:", std_rev)
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Product Category
Books 29597.938776
Beauty 27647.000000
Groceries 11978.000000
Name: Revenue, dtype: float64
Region
East 126232.044039
North 153244.353383
South 104855.736842
West 81197.526316
Name: Revenue, dtype: float64
Month
1 43994.000000
2 33028.000000
3 10557.000000
4 57298.526316
5 27915.000000
6 28190.000000
7 47851.292159
8 76958.000000
9 30719.105263
10 38017.000000
11 27662.736842
12 43339.000000
Name: Revenue, dtype: float64
Mean: 4900.312216631804
Median: 3282.1052631578946
Standard Deviation: 4721.171795257337
PS C:\Users\Inbox\AppData\Local\Programs\Microsoft VS Code>

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